

MARVIN OKONGO

Senior Software Engineer | Technical Lead

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[LinkedIn](#) | [GitHub](#) | [YouTube](#)

SUMMARY

Technical Lead specializing in Golang-based distributed systems and high-performance backend architectures. Expert in building scalable microservices, low-latency APIs, and cloud-native infrastructure. Deeply experienced in Go (Golang) for building concurrent, production-grade systems, complemented by deep experience in Rust for performance-critical components. Pioneer in spec-driven development, leveraging Python's type hinting, validation frameworks (e.g., Pydantic), and structured schemas to minimize LLM hallucinations and ensure reliable, verifiable code generation.

CORE TECHNICAL SKILLS

Primary Stack: Golang (Gin, Fiber, Echo, gRPC, Go Kit, Cobra, Viper)

Supporting Languages: Python (FastAPI, Django), TypeScript/JavaScript, PHP(Laravel)

Mobile: Flutter, React Native, Expo

Databases: MongoDB, PostgreSQL, MySQL, Redis, SQLite

Cloud & DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, AWS (SNS/SQS)

Databases: PostgreSQL, MongoDB, DynamoDB, Vector DBs (Chroma, FAISS)

Architecture: Microservices, Event-Driven Systems, WebAssembly (WASM), TDD

Monitoring: Prometheus, Grafana, OpenTelemetry

SELECTED PROJECTS

Audium Healthcare Management System

Go, Gin, React/Wails, PostgreSQL

[Github](#)

Audium is a multi-portal audiological-healthcare platform (admin, employee, client) that uses a clear layered architecture and domain separation.

- Implemented secure auth foundations with JWT, MFA-ready flows, and role-based controls for sensitive operations.
- Structured service/repository layers to keep business logic testable and improve long-term maintainability.
- Modeled healthcare data with PostgreSQL + GORM and migration/seed tooling for reliable environment setup.

Match Social Platform Backend

Go, Gin, Flutter, PostgreSQL, Redis, S3/MinIO

[Github](#)

Match is a social-discovery and engagement application that powers OTP auth, profiles, discovery/swipes, messaging, reels/stories, and moderation workflows.

- Designed media upload flow using pre-signed URLs and object storage to reduce API payload pressure.
- Added operational safeguards such as rate limits for OTP, swipes, and messages to reduce abuse risk.
- Built reusable response/error patterns to improve frontend integration consistency and supportability.
- Worked with Redis-backed patterns and relational data models to balance speed and transactional correctness.

Mobo Eats

Go, Gin, Flutter, PostgreSQL, WebSockets, Maps

[App Store](#) | [Google Play](#)

Mobo Eats is an all-in-one on-demand commerce platform that lets users order meals, groceries, and nutrient supplements from a single app, with optional web ordering

- Built backend capabilities for merchant/customer flows including storefront onboarding, product/menu management, and order lifecycle tracking.
- Implemented real-time order status updates and tracking-oriented interactions to keep customers and merchants synchronized.
- Supported personalized experiences such as tailored recommendations, diet-focused options, and convenience-first multi-category discovery.

EXPERIENCE

Payserve Limited

Technical Lead (Contract | 05/2024 – Present)

- Build and maintain production services for invoicing, payment workflows, resident operations, and communication pipelines using Go, Nodejs, React, React Native(mobile)
- Design REST APIs and background workers consumed by React and mobile applications, ensuring predictable contracts across teams.
- Implement integrations around M-Pesa/Daraja, email, and SMS channels to support transaction updates and customer engagement.
- Improve service reliability through structured logging, queue-based processing, and better failure handling for async jobs.
- Optimize MongoDB/PostgreSQL queries and indexes to improve consistency of API response times under growing load.

Belva Digital

Software Engineer (12/2021 – 03/2024)

- Built and maintained client-facing applications using Laravel, Go, and Python for campaign and business platforms: Pika na Power, ViuTravel, Co-op Bank, KPLC
- Designed relational database schemas in MySQL/PostgreSQL with attention to performance, normalization, and reporting needs.
- Worked closely with frontend engineers to define API payloads and reduce integration friction during delivery.
- Refactored backend modules to improve maintainability and reduce response-time bottlenecks in core workflows.

Deveint Limited

Software Engineer (05/2021 – 11/2021)

- Built backend services in Go for core business logic and API endpoints used by internal and client-facing products.
- Delivered supporting Python services for selected integrations, utilities, and workflow-specific backend tasks.
- Integrated and maintained APIs consumed by Flutter, React, and Next.js applications with stable data contracts.
- Designed data models and query patterns to improve consistency, correctness, and developer onboarding speed. Collaborated across teams during feature planning, implementation, testing, and production issue resolution.

Soft IQ Technologies Ltd

Software Engineer (03/2021 – 08/2021)

- Developed operational dashboards using React and FastAPI to improve visibility for logistics and internal reporting.
- Implemented authentication and admin features using JWT-based access controls with role-aware dashboard behavior.
- Configured Linux server workflows including SSH deployment routines and Nginx service configuration.
- Assisted with integration work connecting business processes to Microsoft Dynamics 365 and custom APIs.

EDUCATION

- **Full Stack Software Development Certificate** | Moringa School | 12/2023 – 06/2024
- **BSc. Business Information Technology** | Technical University of Kenya | 06/2019 – 12/2023

CERTIFICATIONS

- **Kubernetes and Cloud Native Essentials** — Linux Foundation
- **Mastering Go** — Udemy
- **Docker & Kubernetes Essentials** — Udemy
- **API Programming with FastAPI & Flask** — Udemy

ADDITIONAL STRENGTHS

- **Systems Engineering:** Strong understanding of concurrency, memory efficiency, and high-performance backend design using Go (goroutines, channels, context handling).
- **Platform Engineering:** Experienced in building developer tools, internal frameworks, and CLI utilities (Cobra, Viper) to streamline workflows and reduce boilerplate.
- **Reliability & API Design:** Focused on secure, observable, and production-grade API systems with robust error handling, logging, and monitoring.